

Special advanced lab course A

Quantum State Tomography and HOM Interference

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Abstract

In this experiment you will measure and reconstruct the joint quantum state of polarization-entangled photon pairs. In addition, you will measure the two-photon interference arising when two indistinguishable photons impinge on a beam splitter, known as the Hong-Ou-Mandel (HOM) effect.

1 Introduction

For various applications of quantum physics it is important to reconstruct the quantum state produced by an experiment or apparatus. A fundamental challenge for this reconstruction is that quantum systems are often in superposition states and a measurement causes a collapse of this superposition. This makes it impossible to completely characterise a single quantum system in all possible bases. For example, the orientation of the spin of a particle cannot be measured accurately in all directions at the same time. However, if one has access to multiple copies of a quantum state, one can conduct measurements in various bases and reconstruct the quantum state based on the frequency of the various measurement outcomes. This is essential idea of quantum state tomography (QST) [1]. In quantum photonics, this tomography is done by counting photons emitted from a photon source, but the concepts are transferable to other quantum systems as well.

It is well known that single quanta can exhibit interference effects completely analogous to classical electromagnetic waves. Consider for example single photons or electrons impinging on a double slit one after another. If there is no information available through which of the two slits the particle is propagating, the probability distribution of the particles on a screen behind the slit matches exactly the intensity pattern of light in Young's double slit experiment [2]. However, quantum physics can provide for even more intricate effects: When two or more identical particles are involved one needs to take their bosonic or fermionic exchange symmetry into account. This then allows for interference effects manifesting as *correlations* between the particles. The probably most prominent example is two-particle interference, also known as the Hong-Ou-Mandel (HOM)-effect. It has first been demonstrated in 1987 with photons impinging on a beam splitter [3] and since then been observed in many other quantum systems and extended to more complex scenarios with more particles [4].

2 Theory

2.1 Entangled quantum states

Entanglement is a joint property of quantum systems composed of two or more parts, which cannot be decomposed into individual properties of these parts alone. For concreteness, we consider the polarization of photons as this property throughout the rest of this manual. For a single photon, polarization forms an orthogonal basis as

$$|H\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |V\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

H and V refer to horizontal (0°) and vertical polarizations (90°) of the photons with respect to a suitable frame of reference (e.g. the lab table). An example for an entangled state of the two photons in spatial modes A and B is given by

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}(|H\rangle_A |H\rangle_B + |V\rangle_A |V\rangle_B).$$

If we measure the polarization of photon A and if the outcome is H , then the state of the system will collapse to $|H\rangle_A |H\rangle_B$. Any subsequent measurement in the same basis for photon B will always have H as the outcome, such that the two measurement results are not independent

from another, but correlated. Formally speaking, an entangled state such as $|\phi^+\rangle$ cannot be decomposed into a tensor product of the state of photon A ($|\psi\rangle_A$) and the state of photon B ($|\phi\rangle_B$), a so-called product state ($|\psi\rangle_A \otimes |\phi\rangle_B$), which would produce independent measurement results for its sub-systems:

$$|H\rangle_A |H\rangle_B + |V\rangle_A |V\rangle_B = |H\rangle_A \otimes |H\rangle_B + |V\rangle_A \otimes |V\rangle_B \neq |\psi\rangle_A \otimes |\phi\rangle_B \forall \{|\psi\rangle_A, |\phi\rangle_B\}.$$

This gives rise to the following definition: All states of a composite system that cannot be represented as product states are entangled states.

There is a variety of measures, how much entangled a state is. The following four two-particle states, known as the Bell states, are maximally entangled by all these measures. In the context of photon polarization, they have the following form (one being the state $|\phi^+\rangle$ from above):

$$\begin{aligned} |\psi^+\rangle &= \frac{1}{\sqrt{2}} (|HV\rangle + |VH\rangle) & |\psi^-\rangle &= \frac{1}{\sqrt{2}} (|HV\rangle - |VH\rangle) \\ |\phi^+\rangle &= \frac{1}{\sqrt{2}} (|HH\rangle + |VV\rangle) & |\phi^-\rangle &= \frac{1}{\sqrt{2}} (|HH\rangle - |VV\rangle), \end{aligned}$$

where we omitted the indices A and B for brevity. Additionally, we can express the Bell states in the D (iagonal)/ A (ntidiagonal) basis, which are linear polarizations rotated by $\pm 45^\circ$:

$$|D\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |A\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

The Bell states in this case are as follows:

$$\begin{aligned} |\psi^+\rangle &= \frac{1}{\sqrt{2}} [|DD\rangle - |AA\rangle] & |\psi^-\rangle &= \frac{1}{\sqrt{2}} [|AD\rangle - |DA\rangle] \\ |\phi^+\rangle &= \frac{1}{\sqrt{2}} [|DD\rangle + |AA\rangle] & |\phi^-\rangle &= \frac{1}{\sqrt{2}} [|DA\rangle + |AD\rangle]. \end{aligned}$$

For this basis change the $|\psi^-\rangle$ and the $|\phi^+\rangle$ state keep the same form. In general, if we change the basis for both subsystems the same way, the $|\psi^-\rangle$ or singlet state will always keep its form, where as the other (triplet) states can change into each others' forms. Another relevant set of basis states are R (ight-handed) and L (eft-handed) circular polarisation¹:

$$|R\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad |L\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

Regardless of the particular choice of basis, a polarisation measurement on the two entangled photons will always give correlated results. For example, if we measure one photon of a $|\psi^+\rangle$ -state to be D -polarised, the other photon will be necessarily D -polarised as well. In contrast, for a product state, the two polarisations would be independent. The fundamental idea of quantum state tomography is now to measure the polarisation of the photons in various basis combinations and deduce from the outcome in which state they have been prepared.

¹Which of the two polarisations is assigned i and which $-i$ is a sign convention. Both conventions exist in the literature.

So far the two-photon states have been implicitly assumed as pure states, i.e. a single coherent superposition of the basis states. In reality the polarisation of the photons as emitted does not need to be constant in time or space. Therefore, the polarisation state of the photon pair may also be given as a statistical mixture of pure states, i.e. a mixed state. In general a mixed state can be expressed as an incoherent sum of pure states $|\psi_i\rangle$:

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle \langle\psi_i|,$$

with probabilities p_i with $\sum_i p_i = 1$. For photon pairs, four orthogonal basis states are required. In the HV -basis, these are $\{|HH\rangle, |HV\rangle, |VH\rangle, |VV\rangle\}$, such that the pure states $|\psi_i\rangle$ can be expressed as a superposition of these. Hence, an arbitrary two-photon state (entangled or not) can always be represented as a four-by-four density matrix.

Exercise 1: Work out the density matrix for one of the Bell states above. How does the density matrix change for a mixture consisting of $p_1 = 1/2$ of this Bell state and of $p_2 = 1/2$ of the totally mixed state $1/4I$, with I being the identity matrix?

2.2 Quantum state tomography

The goal of quantum state tomography (QST) is to find a best estimate for an unknown quantum state by measurements. As measurements necessarily disturb a quantum system (in case of photons they even destroy the photons via absorption at the photon detector), QST makes only sense if one has access to multiple copies of the system. In quantum photonics, quantum states are typically produced as a stream of photons (implicitly assuming constant conditions throughout the experiment) and a measurement of the quantum state can be made via measuring the probabilities for certain events: If N photons are detected in total and N_H of them are found to have polarisation H, then the best estimate for the probability of this polarisation is obviously $P_H = N_H/N$.

For single qubit states, which are represented by 2×2 -density matrices, three pairs of measurements in different bases are sufficient to reconstruct the state. This can be seen by representing the density matrix as a superposition of Pauli matrices:

$$\hat{\rho} = \frac{1}{2} \sum_{j=0}^3 S_j \hat{\sigma}_j, \tag{1}$$

with the Pauli matrices $\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ and $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and the Stokes parameters S_j . The Stokes parameters² S_1 to S_3 can be directly obtained from projective measurements into the D/A , R/L , H/V -basis, respectively, whereas S_0 is always

²As in classical optics, the Stokes parameters are the coordinates of the polarisation state on the Poincare-sphere, only that here a different convention of which axis corresponds to which polarisation is used than in most classical optics textbooks.

one due to normalisation [1]. Explicitly, one gets:

$$\begin{aligned} S_0 &= P_H + P_V = 1 \\ S_1 &= P_D - P_A \\ S_2 &= P_R - P_L \\ S_3 &= P_H - P_V. \end{aligned}$$

For photon pairs, one needs to measure both photons of the pair in parallel and record coincidence events for the various polarisation settings of interest. For example, $C(H, D)$ would be the rate of coincidences where the photon in mode A is detected with H polarisation and photon in mode B with D polarisation. Similar to the decomposition into Pauli matrices with Stokes parameters as weights for single qubits as in Eq. 1, for two-qubit states one can decompose the density matrix into tensor products of Pauli matrices with 16 two-qubit Stokes parameters $S_{i,j}$, with $i, j = 0, \dots, 3$. $S_{0,0} = 1$, due to normalisation, which leaves in principle 15 free parameters. One can show that these parameters can be unambiguously reconstructed by measuring the coincidences in the nine possible basis combinations of H/V , D/A and R/L with four settings per combination, i.e. with 36 measurements in total [1]. For example, the parameter $S_{1,1}$ is obtained via measuring both photons in the D/A -basis: $S_{1,1} = P_{D,D} - P_{D,A} - P_{A,D} + P_{A,A}$. For a complete set of equations, see Eqns. (35), (66) and (67) in Ref. [1].

2.3 HOM interference

When a single photon is incident on a balanced beam splitter (see Fig. 1(a)), it will be either reflected or transmitted with a probability of $1/2$. Repeating the experiment many times, one will find on average half the photons in each of the two possible outputs. This recovers the equal splitting of optical intensity, as we know it from classical optics. A closer look on the inner construction of a beam splitter reveals that it also imposes a certain phase on the light, which it reflects or transmits³. This leads to the following transfer matrix for a symmetric beam splitter:

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}, \quad (2)$$

which relates the input and output modes 1 and 2 with each other. The matrix element $U_{i,j}$ gives the probability amplitude for a photon from mode i to propagate to mode j . Hence, a single photon starting in mode 1 will be transformed into a superposition of a photon leaving the beam splitter in port 1 and a photon leaving in port 2: $|\Psi_{\text{out}}\rangle = 1/\sqrt{2}(|1\rangle + i|2\rangle)$.

If now a photon pair is incident on the beam splitter (Fig. 1(b)), with one photon in each of the two input ports ($|\Psi_{\text{in}}\rangle = |1, 2\rangle = |1\rangle \otimes |2\rangle$), a coincidence between the outputs, i.e. an event with exactly one photon in each of the two output modes, can arise either from transmission of *both* photons or from reflection of *both* photons. Hence, the probability for such a coincidence becomes

$$P_{1,1} = |U_{1,1}U_{2,2} + U_{1,2}U_{2,1}|^2 = \frac{1}{4} |1 \cdot 1 + i \cdot i|^2 = 0, \quad (3)$$

see Fig. 1(c). Hence, there are no coincidences and the two photons must always appear together in one of the two outputs. This is a striking behaviour given that there is no force

³dictated by the boundary conditions of the electric field at the interfaces within the beam splitter; several constellations of the phases are possible, all leading to the same result regarding HOM interference

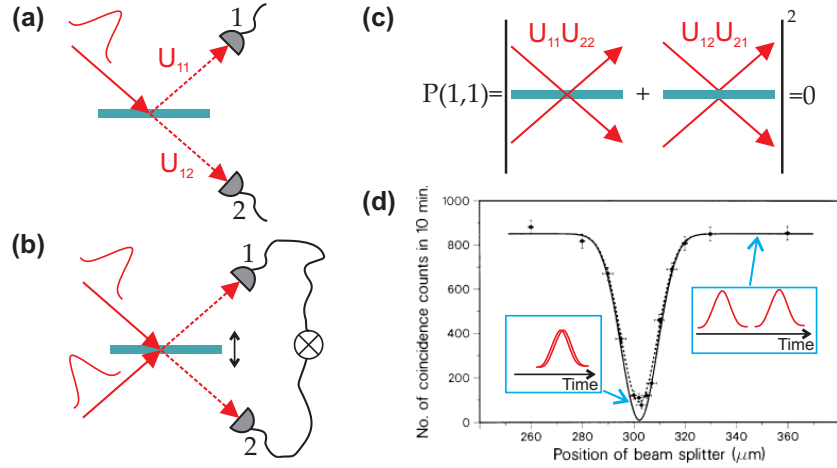


Figure 1: HOM interference at a beam splitter. **(a)** A single photon impinging on a beam splitter with input/output modes 1 and 2 is split into the two probability amplitudes $U_{1,1}$ and $U_{1,2}$. **(b)** Setting of HOM interference: Two photons are incident on the beam splitter and coincidences between the two outputs are recorded. A variable beam splitter position is used to vary the temporal overlap of the photon wavepackets. **(c)** Calculation of the coincidence probability: The two amplitudes for a transmission and reflection of both photons interfere destructively, causing zero coincidences in the ideal case. **(d)** Modified original graph from the HOM experiment [3]. At maximal overlap a minimal number of coincidences is recorded.

between the photons whatsoever, so one might expect a coincidence probability of $1/2$ for independent particles. Nevertheless, a strong statistical correlation between the output modes of the photons arises due to the interference of the two possibilities. This is a two-particle interference effect and there is no classical or single-particle equivalent to it.

It is important to note, however, that the calculation above implicitly assumes *indistinguishability* of the two photons: If the two photons are absolutely identical in all their degrees of freedom, except the spatial mode before the beam splitter, most importantly their polarisation, spectrum and time of arrival on the beam splitter, there is no way to distinguish whether the photon from input mode 1 has been reflected into output mode 1 (with probability amplitude $U_{1,1}$) and the photon 2 reflected into mode 2 ($U_{2,2}$) or vice versa ($U_{1,2}$ and $U_{2,1}$). Therefore, the probability amplitudes are added and interfere destructively, due to the beam splitter phase.

A fundamentally different scenario arises, when the photons are distinguishable, such that one can in principle tell apart which photon went which way. Then the probabilities of the individual events are added instead of the probability amplitudes, and the interference vanishes:

$$P_{1,1,\text{dist}} = |U_{1,1}U_{2,2}|^2 + |U_{1,2}U_{2,1}|^2 = \frac{1}{4}(|1 \cdot 1|^2 + |i \cdot i|^2) = \frac{1}{2}. \quad (4)$$

In their landmark experiment, Hong, Ou and Mandel (HOM) have varied the indistinguishability of two photons by varying the temporal overlap of the photon wavepackets at the beam splitter and observed a continuous transition between these two cases (see Fig.1(d)) [3]. The two-photon interference bears their name since then. For a review of HOM interference, various examples and applications, see Ref. [4].

2.4 Spontaneous parametric down-conversion

In this experiment, we generate photon pairs via spontaneous parametric down-conversion (SPDC) in a nonlinear optical crystal. Depending on the setting of the source, the photons can be polarisation-entangled, distinguishable or indistinguishable. Note that the same source is used in the FPA experiment "Entanglement and Bell's inequality", so this part of the instructions and setup are identical to the other experiment.

SPDC is a nonlinear optical process which relies on the second-order optical nonlinearity of certain media (requiring non-inversion-symmetric crystals); for example, KDP, LiIO_3 , KNbO_3 , LiNbO_3 and BBO. In this experiment, a violet (404 nm) pump laser down-converts into two infra-red fields, known as the signal and idler, inside a BBO (β - BaB_2O_4) crystal. The process is a distinctly quantum mechanical phenomenon, just like the spontaneous emission of an excited atom, and it always produces signal and idler photons in pairs. Two conservation laws have to be obeyed. The first law is the conservation of energy, which requires the frequency of the pump photon to equal the sum of the frequencies of the two generated photons.

$$\hbar\omega_{\text{pump}} = \hbar\omega_{\text{signal}} + \hbar\omega_{\text{idler}}$$

Furthermore, the photons are emitted diametrically opposite to one another around the pump direction, which is a consequence of the conservation of momentum. It requires the wavevector of the pump photon to equal the sum of the wavevectors of the two generated photons.

$$\hbar\mathbf{k}_{\text{pump}} = \hbar\mathbf{k}_{\text{signal}} + \hbar\mathbf{k}_{\text{idler}} \quad (5)$$

In general it is difficult to satisfy the momentum conservation, because all media are dispersive and the wavenumber is proportional to the frequency-dependent refractive index $n(\omega)$, i.e. $k(\omega) = \omega n(\omega)/c$. If the momentum conservation is not fulfilled, signal (and idler) photons created in different locations of the crystal acquire a different phase and interfere destructively with each other. The photons are produced in phase (one also speaks of "phase-matching") only if (5) is satisfied. In our setup, we exploit the birefringence of the BBO crystal to achieve phase-matching. Specifically, the signal photon carries extraordinary polarization (here V), while the idler photon is ordinarily polarized (here H). This is called type-II phase-matching. As a consequence, signal and idler photons are emitted on two distinct cones, as can be seen in Figure 2.

Polarization entangled photons can be obtained by spatial filtering of the photons emerging on the intersection lines of the two cones and spectral filtering around the degeneracy frequency $\omega_{\text{signal}} = \omega_{\text{idler}} = \omega_{\text{pump}}/2$. Momentum conservation guarantees that one photon will emerge at each intersection, but it will be impossible to distinguish which cone either photon came from. However, the two photons must carry opposite polarizations.

Exercise 2: What kind of two-photon state do you expect at the intersection of the cones and why?

3 Experimental setup

Figure 3 shows the schematic setup of the photon pair source and optical components needed for the experiment. The pump laser is focused into the BBO crystal, where photon pairs are

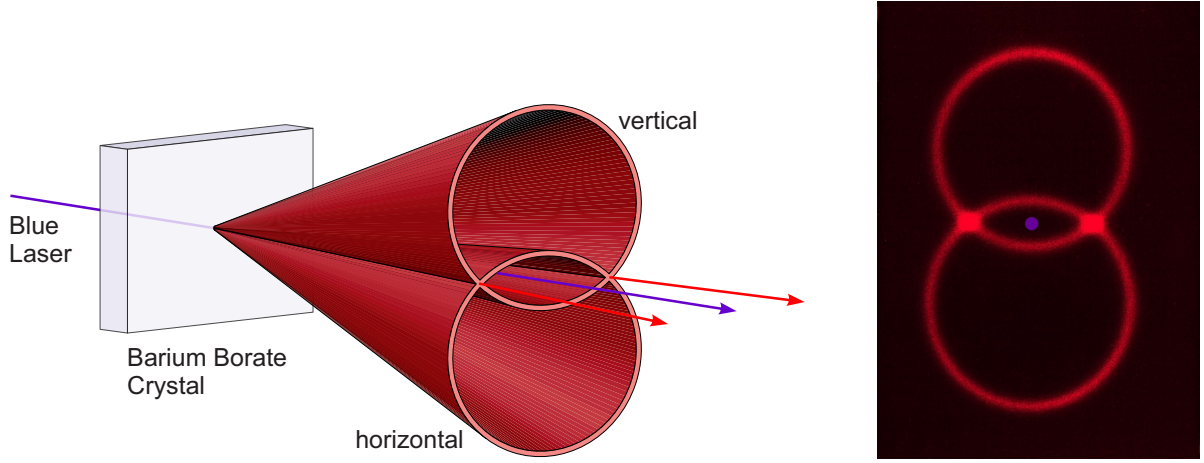


Figure 2: Left: Type-II SPDC produces photon pairs, which emerge on two cones where the vertically polarized photon is on the upper cone and the horizontally polarized photon is on the lower cone. Polarization entangled photons are observed at the two intersection lines of the cones. Right: False-color photograph of the down-conversion light taken with a bandpass filter at the degenerate wavelength.

produced via SPDC. The intersection directions of the cones are separated by two small prism mirrors and then directed via mirrors (not shown) and further lenses into optical fibers to eventually detect the two photons of a pair. Due to the birefringence of the BBO crystal the propagation directions and velocities of horizontally and vertically polarized light are slightly different as they travel through the crystal. This leads to a transverse walk-off (spatial displacement) and longitudinal walk-off (temporal displacement) of the horizontally polarized photons from the vertically polarized photons. Thus the photons can be distinguished by their position and relative timing and, thus, do not contribute to the entanglement. To preserve the polarization entanglement, we will have to make sure that there is no information available that can be used to determine the polarization state of the photons. So, it is necessary to compensate the walk-off before the measurements. While this cannot be done completely, the distinguishability of the photons can be much reduced by the use of compensator BBO crystals. First, the down-converted photons have their polarizations rotated by 90° using a halfwave plate so that H and V get exchanged. Each photon is then passed through a compensator BBO crystal which is half as thick as the original crystal that performs down-conversion. In this fashion, the walk-off effects are reversed by half, which consequently erases all temporal distinguishability and reduces the effects of the transverse walk-off. The temporal compensation is illustrated in Figure 4.

In the setup there are additional spectral filters and polarizers. The filters remove stray pump light and ensure that only photons of (nearly) the same wavelength are coupled into the fibers (the bandpass filters have 3 nm bandwidth). The polarizers can be inserted to perform the correlation measurements. QST can be performed by rotating the waveplates behind the compensation crystal to the angles needed to project the state into desired bases (H/V , D/A and R/L). The fibers are finally connected to single photon detectors (APD - avalanche photodiode), which trigger a macroscopic current when a photon impinges on them. The APDs are

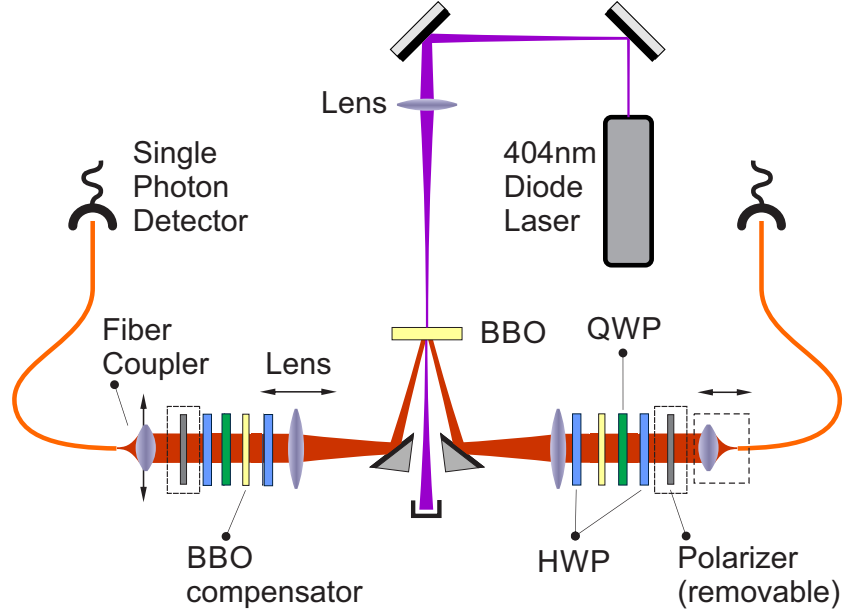


Figure 3: Schematic of the experimental setup. A blue laser pumps the BBO crystal to create photon pairs. The intersection lines of the H and V cones deviate from the pump beam by about 3° . Small prism mirrors separate the down-conversion beams, which pass a collimating lens, a half-wave-plate (HWP), the compensating BBO crystal, a further combination of quarter-wave plate (QWP) and HWP as well as a removable polarizer, before getting coupled into a single-mode optical fiber for spatial selection and eventually detection. The actual setup includes color glass filters and bandpass filters for spectral selection as well as a fiber beam splitter for the HOM experiment (not shown here).

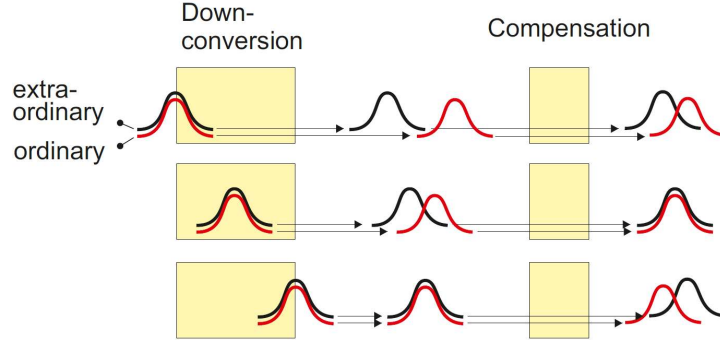


Figure 4: Walk-off compensation by reversing half of the relative time delay between ordinary and extraordinary photons of a pair. In this way the relative time delay reveals no information about the polarization state of a photon.

connected to a logical circuit, which counts current pulses from both detectors arriving within a time-window of 8 ns as a coincidence and displays the coincidence rate on a computer.

Exercise 3: Given the waveplates have their fast axes oriented at 0° , to which angles do you need you turn the components in order to measure the quantum state in the H/V , D/A and R/L bases, respectively?

For the HOM experiment, an in-fiber beam splitter (not shown here) can be inserted between the two collection fibers and the detectors. The indistinguishability can be adjusted via translating one of the fiber collection optics along the beam propagation direction (see rightmost arrow in Fig. 3), which varies the temporal overlap of the photon wavepackets.

Exercise 4: What is the expected length of the photon wavepackets? This determines how precisely one has to set the position of the fiber.

4 Experimental procedure

4.1 Laser safety



The pump laser has a wavelength of 404 nm and operates at a power of 50 mW, which places the laser in class 3B. This means that a direct hit of the laser beam to the eye (including reflections from a mirror or any other shiny surface) is likely to cause permanent damage to the retina. However, as opposed to class 4 lasers, diffuse stray light from matt surfaces is safe. The photon flux from the near-infrared photons is also too weak to cause any harm. Your instructor will give you a laser safety briefing at the start of the experiment. Here is a summary of the most important rules, which you must respect at all times when working with the experiment:

1. Under no circumstances look directly into the pump laser beam path! The pump laser is mostly contained in one half of the setup and enters the other only to pass the crystal before it is absorbed in a beam dump (see Fig. 3).
2. All optical elements (mirrors, lenses, crystal, etc.) are attached to the table. This ensures that the laser beam does not leave its designed path. Do not loosen or shift any element! A tumbling mirror may cause uncontrollable reflections anywhere in the lab. Use only the fine-adjustment screws for alignments, if necessary.
3. Do not wear watches, jewelry or any other shiny surfaces on your hand or wrist! They may just as well cause uncontrollable reflections.
4. There may be other experiments running in the lab at the same time. The scientists working on these are informed of your presence and will make sure that no harmful radiation enters the area where you are working in. In any case, do not enter other zones of the lab on your own and do not stick your head above or into other optical setups (obviously do not touch anything either), as dangerous laser radiation might be present in those spaces.

4.2 Quantum state tomography of a Bell state

The first task for this experiment is to perform a quantum state tomography of the polarisation state as emitted by the source. For this you need to measure the coincidence rates in 9 different basis and at four polariser settings in each basis, as discussed in section 2.2. For ease of notation we refer to the polariser settings as $++$, $+-$, $-+$ and $--$. The notation $++$ means both polarisers are not rotated with respect to the specified basis setting, i.e. in the

D/D -basis, we would measure the coincidence rate $C(D/D)$. The setting $+-$ then corresponds to polariser B being rotated by 90° (measuring $C(D/A)$), and so on. So you need to fill a table of the form (see also Eq. (66) in Ref. [1]):

Bases (A/B)/ Combination	++	+-	-+	--
D/D				
D/R				
D/H				
R/D				
R/R				
R/H				
H/D				
H/R				
H/H				

Table 1: Matrix of required measurements for QST.

Then you can reconstruct the density matrix of the state using Eq. (35) from Ref. [1]. You may also find the code provided at <https://research.physics.illinois.edu/QI/Photonics/tomography/> useful for your analysis.

For many applications in quantum information processing it is desirable to produce a state as close as possible to one of the four maximally entangled Bell states. Adjust the compensation BBO crystals to produce either the $|\psi^-\rangle$ or the $|\psi^+\rangle$ state. Then redo the QST procedure and reconstruct the density matrix of this state. Calculate its fidelity to the ideal Bell state and discuss possible limitations.

4.3 HOM interference

The second task is to demonstrate the HOM effect. For this purpose you should adjust the polarisation of the two photons to be non-entangled and indistinguishable. Then attach the fiber beam splitter to the two collection fibers and attach its outputs to the detectors. Slowly vary the delay stage of one of the fiber collimators to change the optical path length. Search for a region where there is a strong drop of the coincidence rate. This could be the region where the temporal wavepackets overlap. Run a fine scan around this point and record the HOM interference pattern. Calculate the visibility of the curve and discuss its limitations.

If there is enough time, you can also try to repeat the measurement with polarisation-distinguishable photons as well as with entangled photons in the $|\psi^-\rangle$ and $|\psi^+\rangle$ -state, respectively. Analyse and discuss any striking differences you observe.

4.4 Data processing and results

The required data to present in your lab report are:

1. Coincidence data and reconstructed density matrix for the initial state of the source
2. Coincidence data and reconstructed density matrix for an optimised Bell state. Fidelity of this state with the ideal Bell state.
3. Coincidence curve as a function of delay, showing the HOM-interference pattern and its visibility.
4. (optional) HOM-coincidence curve with distinguishable and polarisation-entangled photons.

References

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Appendix - Alignment

The experiment is pre-aligned by the instructor, because it would take too much time to do this from scratch in one day. It is strongly advised not too change the alignment unless the measurements do not yield sensible results (please consult with your instructor, in any case). The alignment proceeds as follows (assuming that you already receive some counts on the APDs):

Initially we align for the highest single count rate for both arms (polarisers removed) by adjusting the pointing of the fiber coupler and its preceding mirror as well as by adjusting the focus of the collimating lens. Second one checks with a polarizer, whether the count rate for H and V polarizations are similar. If they differ by much, more alignment is required. This may reduce the overall count rate a bit. As a next step we monitor the coincidence rate and try to maximize it again by adjusting the mirror, the fiber couplers and the coupling lenses on both sides. Once we have a useful number of coincidence counts (a few hundred per second) we check how the H - V and V - H coincidence count rates compare. They should be roughly equal. If not, more alignment is required. In particular we may need to “walk the beam” by using a small transverse movement on one side (with the appropriate angular adjustment) that is “compensated” by a small movement on the other side. If the H - V and the V - H coincidence count rates are equal to within 20%, we can check the H - H and the V - V coincidence count rates. They should be extremely small. If not, alignment of the first HWPs might be necessary. After that the polarizers should be set to D - D and/or A - A to optimize the compensation crystal angle (by minimising the coincidences). By adjusting the compensation

crystal vertical tilt angle we can set the phase of the state to be $|\psi^-\rangle$ or $|\psi^+\rangle$. Please do not change the alignment of the pump laser beam path or the orientation of the primary BBO crystal. Realignment those would take a lot of time.